

SOFTWARE ENGINEERING-01

Requirements Table

Dropshipping Inventory & Order Orchestrator

Problem Statement #38 | Retail, E-Commerce & Finance | Actors: E-Store Owner, Supplier Partner

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CLASS: D

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FR-001

Type: Functional

Description: The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory in real time, and auto-dispatch supplier purchase orders.

Priority: High

Acceptance Criteria: Pass: Supplier PO triggered with correct customer delivery address. Fail: Order placed for an out-of-stock supplier SKU.

Rationale: Automating webhook ingestion and PO dispatch removes manual order-processing delay and prevents overselling out-of-stock supplier items.

FR-002

Type: Functional

Description: The system shall split a multi-vendor customer order into separate sub-orders, one per Supplier Partner, based on SKU ownership.

Priority: High

Acceptance Criteria: Pass: A checkout with SKUs from 2+ suppliers generates

one PO per supplier with correct line items. Fail: A sub-order is missing items or routed to the wrong supplier.

Rationale: Dropshipping orders routinely span multiple suppliers; without splitting, a supplier could receive or be billed for items it doesn't own.

FR-003

Type: Functional

Description: The system shall subscribe to each Supplier Partner's shipment/tracking webhook or API and attach the tracking number to the corresponding sub-order once available.

Priority: Medium

Acceptance Criteria: Pass: Tracking number appears on the Store Owner's order view within 1 business day of supplier dispatch. Fail: Sub-order remains untracked after the supplier confirms shipment.

Rationale: End-to-end shipment visibility is core to the "Order Orchestrator" promise and is required for customer support and dispute resolution.

FR-004

Type: Functional

Description: The system shall let a Supplier Partner manually update the status of an assigned sub-order (Accepted / Backordered / Shipped / Cancelled) through an authenticated interface.

Priority: Medium

Acceptance Criteria: Pass: Status change is reflected on the Store Owner's dashboard within seconds of submission. Fail: Status update is lost, delayed beyond SLA, or applied to the wrong order.

Rationale: Suppliers need a channel to report exceptions the automated pipeline can't detect on its own (e.g. warehouse damage).

FR-005

Type: Functional

Description: The system shall re-verify supplier stock at the moment of PO dispatch and automatically cancel/refund an affected line item if a previously in-stock SKU becomes unavailable before dispatch completes.

Priority: High

Acceptance Criteria: Pass: A stock change between webhook receipt and

dispatch results in that line item being cancelled, not dispatched. Fail: PO is dispatched for a SKU confirmed out-of-stock at dispatch time.

Rationale: Real-time supplier inventory can change within seconds; a dispatch-time re-check closes the race condition FR-001's own "Fail" case describes.

NFR-001

Type: Nonfunctional (Performance & Security)

Description: Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.

Priority: High

Acceptance Criteria: Pass: Benchmarking confirms target latency and security standards under simulated peak load.

Rationale: Slow acknowledgements cause storefront platforms to retry-flood the webhook endpoint, risking cascading failure.

NFR-002

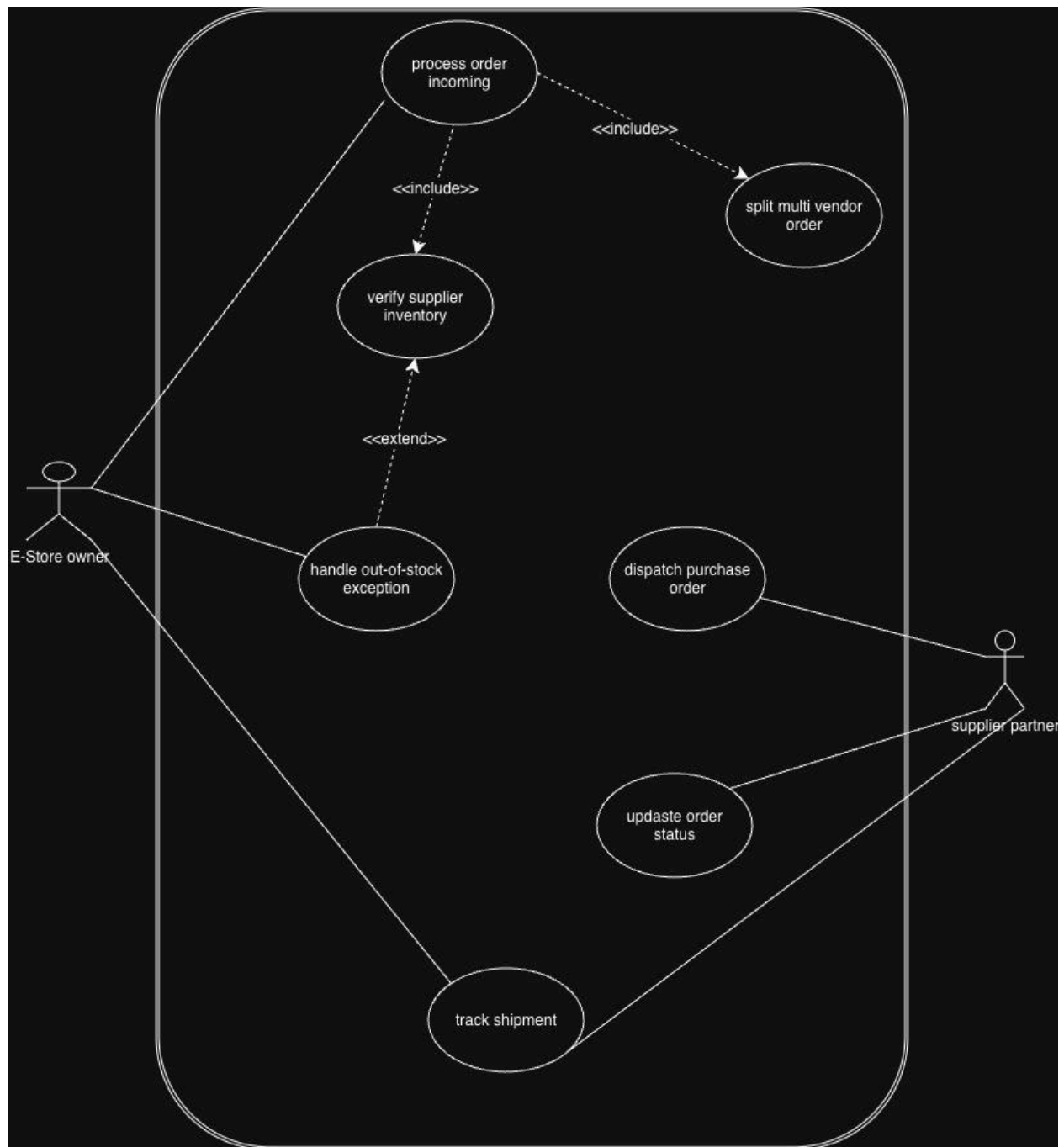
Type: Nonfunctional (Reliability & Data Integrity)

Description: The system shall guarantee idempotent, at-least-once processing of every order webhook so no customer order is dropped or double-processed if the same webhook is redelivered.

Priority: High

Acceptance Criteria: Pass: Sending an identical webhook payload twice results in exactly one PO per supplier. Fail: Duplicate delivery creates a duplicate PO or silently drops the order.

Rationale: Webhook providers commonly redeliver on any timeout or error; idempotency prevents double-billing suppliers or losing customer orders.



Use-Case Flow Specification

Problem Statement #38 — Dropshipping Inventory & Order Orchestrator

Use Case: Process Incoming Order

Primary Actor: E-Store Owner

Secondary Actor: Supplier Partner

Includes: Verify Supplier Inventory, Split Multi-Vendor Order, Dispatch Purchase Order

Extended by: Handle Out-of-Stock Exception

Preconditions

- The storefront (Shopify/WooCommerce) is registered and its order webhook is authenticated with the Orchestrator.
- At least one Supplier Partner has a valid, reachable inventory API/webhook connection.
- A customer has completed checkout on the storefront.

Main Success Scenario

1. The storefront sends an order-created webhook to the Orchestrator.
2. The system acknowledges the webhook with HTTP 200 within 200 ms.
3. The system parses the order details, including line items, SKUs, and customer delivery address.
4. The system splits the multi-vendor order by identifying the Supplier Partner responsible for each SKU.
5. The system verifies real-time supplier inventory for each supplier.
6. The system creates a separate purchase order for each Supplier Partner with the correct SKUs, quantities, and delivery address.
7. The system dispatches each purchase order to the respective Supplier Partner.
8. The system records the supplier purchase orders and marks the customer order as "Processing".
9. The Supplier Partner provides shipment and tracking information.
10. The system attaches the tracking number to the corresponding sub-order and updates the order status.
11. The E-Store Owner can view the updated shipment and tracking information.
12. The use case ends successfully.

Alternate Flow

5a. Supplier Stock Insufficient

5a1. The system detects that a requested SKU or quantity is unavailable from a Supplier Partner.

5a2. The system triggers the **Handle Out-of-Stock Exception** use case.

5a3. The system does not dispatch a purchase order for the unavailable SKU.

5a4. The system notifies the E-Store Owner about the unavailable item.

5a5. The affected item is marked as unavailable, cancelled, or backordered.

5a6. If other items are available, processing continues for those items; otherwise, the affected supplier sub-order is not dispatched.

Postconditions

- Separate supplier purchase orders exist for the applicable Supplier Partners.
- No purchase order is dispatched for an item confirmed as out of stock.
- The customer order status is updated and visible to the E-Store Owner.
- Shipment tracking information is attached to the corresponding sub-order when provided by the Supplier Partner.

Requirements Table — Dropshipping Inventory & Order Orchestrator

Problem Statement #38 | Retail, E-Commerce & Finance | Actors: E-Store Owner, Supplier Partner

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory in real time, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO triggered with correct customer delivery address. Fail: Order placed for an out-of-stock supplier SKU.	Automating webhook ingestion and PO dispatch removes manual order-processing delay and prevents overselling out-of-stock supplier items.
FR-002	Functional	The system shall split a multi-vendor customer order into separate sub-orders, one per Supplier Partner, based on SKU ownership.	High	Pass: A checkout with SKUs from 2+ suppliers generates one PO per supplier with correct line items. Fail: A sub-order is missing items or routed to the wrong supplier.	Dropshipping orders routinely span multiple suppliers; without splitting, a supplier could receive or be billed for items it doesn't own.
FR-003	Functional	The system shall subscribe to each Supplier Partner's shipment/tracking webhook or API and attach the tracking number to the corresponding sub-order once available.	Medium	Pass: Tracking number appears on the Store Owner's order view within 1 business day of supplier dispatch. Fail: Sub-order remains untracked after the supplier confirms shipment.	End-to-end shipment visibility is core to the "Order Orchestrator" promise and is required for customer support and dispute resolution.
FR-004	Functional	The system shall let a Supplier Partner manually update the status of an assigned sub-order (Accepted / Backordered / Shipped / Cancelled) through an authenticated interface.	Medium	Pass: Status change is reflected on the Store Owner's dashboard within seconds of submission. Fail: Status update is lost, delayed beyond SLA, or applied to the wrong order.	Suppliers need a channel to report exceptions the automated pipeline can't detect on its own (e.g. warehouse damage).
FR-005	Functional	The system shall re-verify supplier stock at the moment of PO dispatch and automatically cancel/refund an affected line item if a previously in-stock SKU becomes unavailable before dispatch completes.	High	Pass: A stock change between webhook receipt and dispatch results in that line item being cancelled, not dispatched. Fail: PO is dispatched for a SKU confirmed out-of-stock at dispatch time.	Real-time supplier inventory can change within seconds; a dispatch-time re-check closes the race condition FR-001's own "Fail" case describes.
NFR-001	Nonfunctional (Performance & Security)	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load.	Slow acknowledgements cause storefront platforms to retry-flood the webhook endpoint, risking cascading failure.
NFR-002	Nonfunctional (Reliability & Data Integrity)	The system shall guarantee idempotent, at-least-once processing of every order webhook so no customer order is dropped or double-processed if the same webhook is redelivered.	High	Pass: Sending an identical webhook payload twice results in exactly one PO per supplier. Fail: Duplicate delivery creates a duplicate PO or silently drops the order.	Webhook providers commonly redeliver on any timeout or error; idempotency prevents double-billing suppliers or losing customer orders.

Use-Case Flow Specification

Problem Statement #38 — Dropshipping Inventory & Order Orchestrator

Use Case: Process Incoming Order

Primary Actor: E-Store Owner

Secondary Actor: Supplier Partner

Includes: Verify Supplier Inventory, Split Multi-Vendor Order, Dispatch Purchase Order

Extended by: Handle Out-of-Stock Exception (extends Verify Supplier Inventory)

Preconditions

- The storefront (Shopify/WooCommerce) is registered and its order webhook is authenticated with the Orchestrator.
- At least one Supplier Partner has a valid, reachable inventory API/webhook connection.
- A customer has completed checkout on the storefront.

Main Success Scenario

1. The storefront sends an order-created webhook to the Orchestrator.
2. The system acknowledges the webhook with HTTP 200 within 200 ms (NFR-001).
3. The system parses the payload, extracting line items, SKUs, and the delivery address.
4. The system includes Split Multi-Vendor Order to group line items by owning Supplier Partner.
5. For each supplier group, the system includes Verify Supplier Inventory to confirm real-time stock via the supplier's API.
6. The system generates one purchase order per Supplier Partner, containing only that supplier's line items and the delivery address.
7. The system includes Dispatch Purchase Order to send each PO to its respective Supplier Partner.
8. The system marks the order "Processing" and notifies the E-Store Owner via the dashboard. Use case ends successfully.

Alternate Flow

5a. Supplier Stock Insufficient

5a1. The system detects that a requested SKU/quantity is unavailable for a given supplier during step 5.

5a2. The system triggers Handle Out-of-Stock Exception (extends Verify Supplier Inventory) for the affected line item(s).

5a3. The system cancels/flags the line item, notifies the E-Store Owner of the shortfall, and offers the customer a refund or backorder choice.

5a4. If in-stock items remain for that supplier, flow resumes at step 6 with those items only; otherwise that supplier's sub-order is skipped and flow resumes at step 6 for the remaining suppliers.

Postconditions

- One sub-order/PO exists per involved Supplier Partner, each with the correct SKUs and delivery address.
- The order's status is visible on the E-Store Owner's dashboard as "Processing" or a later state.
- Every dispatched PO corresponds to inventory confirmed available at the moment of dispatch.